BiCross-UNet: Hierarchical Bidirectional Cross-Conditioning and Privileged Denoising Distillation for Robust Perceptive Locomotion

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***Abstract*— Perceptive locomotion is important for the practical deployment of humanoid robots on challenging terrain, where the control policy must effectively exploit terrain observations while remaining robust to noisy and imperfect perception. In this paper, we propose *BiCross-UNet*, a robust perceptive locomotion framework for challenging terrain. Our method jointly encodes proprioceptive history and terrain observations through hierarchical bidirectional cross-conditioning and global cross-attention fusion, aiming to better model body–terrain interactions. To improve robustness under corrupted terrain observations, we further introduce a teacher-guided terrain reconstruction objective, which encourages the student to learn control-oriented terrain representations from noisy inputs. In addition, we develop a stable distillation pipeline that integrates Dataset Aggregation (DAgger)-style action mixing, replay-based optimization, staged training, and a structured corruption curriculum. Experiments and ablation studies demonstrate that the proposed method enables more reliable perceptive locomotion over challenging terrain and achieves improved robustness under uneven terrain perception, while retaining a lightweight architecture suitable for real-time deployment.**

Keywords— Perceptive locomotion, humanoid robots, challenging terrain, teacher–student distillation, body–terrain interaction, robust terrain perception.

I. INTRODUCTION

Perceptive locomotion is essential for the practical deployment of humanoid robots on challenging terrain, where the control policy must effectively exploit terrain observations while remaining robust to noisy and imperfect perception. Compared with flat-ground locomotion, challenging terrain—such as discontinuities, sparse stepping regions, narrow supports, deformable surfaces, and irregular height variations—requires the robot not only to maintain stable whole-body motion, but also to anticipate terrain geometry and adapt its motion accordingly. To support such terrain-aware control, robot-centric elevation maps and related local terrain representations are widely used as compact descriptions of surrounding geometry. However, in practical deployment, terrain perception is rarely ideal. Elevation maps may contain noise, bias, drift, dropouts, and spatially correlated errors, which can mislead the controller and significantly degrade locomotion performance.

Recent advances in reinforcement learning (RL) have led to substantial progress in agile legged and humanoid locomotion. For perceptive locomotion, one important line of work focuses on extracting action-relevant terrain information from exteroceptive observations. For example, attention-based map encoding conditions terrain feature selection on the robot state and command, thereby improving terrain-aware control on foothold-critical terrain [1]. Another line of work improves deployability through teacher–student distillation, where a teacher policy trained with privileged simulation information transfers knowledge to a student policy using only onboard observations, thereby improving robustness and sample efficiency under noisy exteroception [2]. Despite these advances, current methods still face two major limitations. First, body and terrain information are often fused only once, such as through feature concatenation or a single cross-attention stage, which limits the policy's ability to model body–terrain interactions across multiple representation levels. Second, robustness to imperfect terrain perception is often addressed by generic noise injection or stochastic latent modeling, which may not adequately capture realistic corruption in terrain observations and may also complicate optimization.

To address these limitations, we propose *BiCross-UNet*, a robust perceptive locomotion framework for challenging terrain. Our method jointly encodes proprioceptive history and terrain observations using a two-stream architecture, where proprioceptive dynamics are modeled by a recurrent encoder and terrain observations are processed by a lightweight hierarchical terrain encoder. Instead of fusing

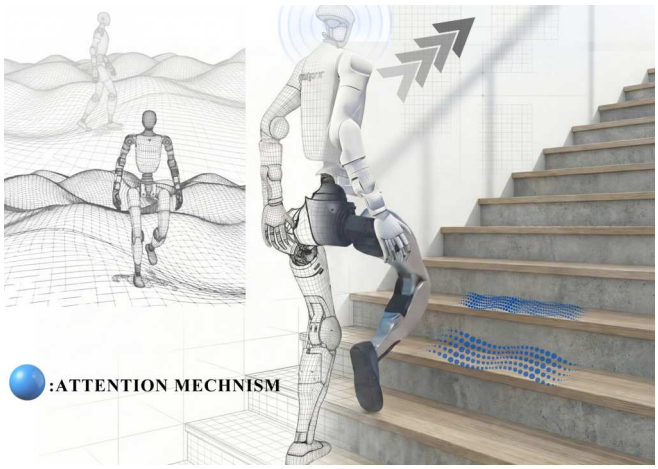


Fig. 1. Simulation and real-world experiments of perceptive locomotion in a structured stair-like scenario. The figure shows the terrain-aware representation used for control together with the corresponding real-world deployment. The blue region indicates the perceived terrain area.

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the two modalities only at the final stage, we introduce *hierarchical bidirectional cross-conditioning*, which allows body features to guide terrain encoding and terrain features to refine body representations throughout the encoding hierarchy. A global cross-attention fusion module is then used to produce control-oriented terrain representations for locomotion. In this way, the policy learns a tighter coupling between body dynamics and terrain structure, which is important for reliable locomotion on non-flat and foothold-sensitive terrain.

To further improve robustness under corrupted terrain observations, we introduce a *teacher-guided terrain reconstruction objective*. During training, the student receives noisy terrain observations, while teacher-guided supervision encourages the latent terrain representation to preserve geometric information that is useful for locomotion control. This auxiliary objective complements action distillation by explicitly regularizing the terrain encoder toward more reliable control-oriented representations. In addition, we develop a stable distillation pipeline that integrates Dataset Aggregation (DAgger)-style action mixing, replay-based optimization, staged training, and a structured corruption curriculum. These designs improve optimization stability and sample efficiency, while enabling the student policy to gradually adapt from privileged supervision to deployment-oriented observations.

In summary, our main contributions are as follows:

- We propose *BiCross-UNet*, a perceptive locomotion framework that models body–terrain interactions through hierarchical bidirectional cross-conditioning and global cross-attention fusion, enabling tighter coupling between proprioceptive dynamics and terrain representations on challenging terrain.
- We introduce a teacher-guided terrain reconstruction objective that improves the robustness of terrain representations learned from noisy observations and complements action distillation for perceptive locomotion on challenging terrain.
- We develop a stable distillation pipeline with DAgger-style action mixing, replay-based optimization, staged training, and a structured corruption curriculum, and validate the proposed framework through experiments and ablation studies on challenging-terrain perceptive locomotion.

II. RELATED WORK

A. Learning-Based Perceptive Locomotion on Challenging Terrain

Recent advances in reinforcement learning have substantially improved agile locomotion over rough ground, stairs, and other challenging terrains for legged robots [3]–[5]. Building on this foundation, perceptive locomotion methods incorporate exteroceptive terrain observations into policy learning so that the robot can anticipate footholds and adapt its motion before contact. For quadrupeds, Miki *et al.* [6] learned a robust perceptive controller that filters unreliable exteroceptive inputs in the wild, while DTC [7] combined model-based guidance with a learned tracking controller to improve precise traversal on foothold-critical terrains. More recently, attention-based map encoding [1] showed that conditioning terrain encoding on robot state improves generalized traversal across diverse discontinuous terrains. For humanoids, Perceptive Internal Model [8] and BeamDojo [9] further demonstrated the effectiveness of terrain-centric local representations and sparse-foothold training for challenging locomotion. However, many existing methods still fuse body and terrain information only once, *e.g.*, through direct concatenation or a single attention stage, which limits the ability of the policy to build rich body–terrain interactions across multiple representation levels.

B. Robust Perceptive Locomotion under Imperfect Terrain Perception

A second line of research focuses on robustness when terrain perception is noisy, incomplete, or unreliable. Beyond direct terrain-aware control, latent adaptation and implicit terrain imagination have been explored to reduce dependence on clean exteroception, as exemplified by DreamWaQ [10]. For humanoids, denoising world model learning [11] and the recent teacher–student framework in [2] explicitly address locomotion over challenging terrain under imperfect perception by exploiting privileged information during training. These methods significantly improve deployability, but they typically rely on generic corruption randomization, latent denoising, or one-shot body–terrain fusion, and do not explicitly enforce a control-oriented terrain representation through hierarchical interaction and teacher-guided reconstruction.

More broadly, reliable geometric representations are also central to terrain-aware autonomy beyond humanoid locomotion, including traversability assessment [12], uneven-terrain planning for car-like robots [13], [14], planning for articulated vehicles in unstructured environments [15], safety-critical control based on geometric obstacle abstractions [16], route planning on structured vector maps [17], and feed-forward 3D reconstruction or dense SLAM pipelines [18], [19]. While these works address different autonomy problems, they collectively highlight the importance of robust geometry-aware representations for downstream decision-making. In contrast, our work focuses on perceptive humanoid locomotion on challenging terrain and addresses the above limitations by jointly modeling hierarchical body–terrain interaction and teacher-guided terrain reconstruction within a stable distillation framework.

III. METHOD

We propose *BiCross-UNet*, a teacher–student perceptive locomotion framework for challenging terrain under noisy and imperfect terrain perception. The proposed method consists of three components: 1) a hierarchical body–terrain representation architecture, 2) a teacher-guided terrain reconstruction objective, and 3) a stable distillation pipeline for learning under corrupted terrain observations.

A. Problem Formulation

We formulate perceptive locomotion as a partially observed control problem. At time step t, the student policy receives a proprioceptive history and a terrain observation,

$$o_t = (x_{t-H+1:t},\, \hat{e}_t)\,, \tag{1}$$

where $x_{t-H+1:t} = \{x_{t-H+1}, \ldots, x_t\}$ denotes recent proprioceptive observations, and $\hat{e}_t$ denotes the terrain observation derived from the local terrain representation. In practical deployment, $\hat{e}_t$ may be corrupted by sensing noise, drift, bias, or partial observability.

During training, a teacher policy provides additional supervision through teacher observations o_t^{T} and teacher actions $a_t^{\mathrm{T}} = \pi_\phi(o_t^{\mathrm{T}})$. The student policy,

$$a_t = \pi_\theta(o_t), \tag{2}$$

is trained to produce locomotion actions using only deployment-oriented observations. The goal is to learn a student policy that remains effective on challenging terrain under imperfect terrain perception.

B. BiCross-UNet Architecture

BiCross-UNet adopts a two-stream architecture that jointly models body dynamics and terrain observations. The body stream summarizes short-term motion history, while the terrain stream extracts hierarchical terrain features. Instead of fusing the two modalities only once at the final stage, the proposed architecture couples them repeatedly during encoding.

1) Body stream: The proprioceptive history is encoded by a recurrent body encoder,

$$b_t = f_{\mathrm{body}}(x_{t-H+1:t}), \tag{3}$$

where $b_t \in \mathbb{R}^{d_b}$ is the latent body representation. In our implementation, $f_{\mathrm{body}}(\cdot)$ is realized by a Gated Recurrent Unit (GRU), which captures short-horizon motion dependencies relevant to terrain-aware locomotion.

2) Terrain stream: The terrain observation is first projected into a latent space,

$$h_t^{(0)} = f_{\mathrm{proj}}(\hat{e}_t), \tag{4}$$

and then processed by a hierarchy of terrain encoder blocks,

$$h_t^{(\ell)} = f_{\mathrm{enc}}^{(\ell)}(h_t^{(\ell-1)}), \qquad \ell = 1, \ldots, L, \tag{5}$$

where $h_t^{(\ell)}$ denotes the terrain feature at level ℓ. The final terrain bottleneck representation is

$$z_t = f_{\mathrm{bot}}(h_t^{(L)}). \tag{6}$$

This terrain branch provides the feature representation used for both action prediction and terrain reconstruction.

3) Hierarchical bidirectional cross-conditioning: A key component of BiCross-UNet is *hierarchical bidirectional cross-conditioning*, which models body–terrain interactions throughout the encoding hierarchy. At each interaction level, terrain features are updated using body context, while the body feature is refined using terrain information. Specifically, the terrain-updating branch is

$$\Delta h_t^{(\ell)} = \mathrm{MHA}\Big(W_{Q,h}^{(\ell)} h_t^{(\ell)}, W_{K,b}^{(\ell)} b_t, W_{V,b}^{(\ell)} b_t\Big)\,, \tag{7}$$

and the body-updating branch is

$$\Delta b_t^{(\ell)} = \mathrm{MHA}\Big(W_{Q,b}^{(\ell)} b_t, W_{K,h}^{(\ell)} h_t^{(\ell)}, W_{V,h}^{(\ell)} h_t^{(\ell)}\Big)\,. \tag{8}$$

The updated features are obtained by residual fusion and layer normalization,

$$h_t^{(\ell)} \leftarrow \mathrm{LN}\Big(h_t^{(\ell)} + \Delta h_t^{(\ell)}\Big)\,, \tag{9}$$

$$b_t \leftarrow \mathrm{LN}\Big(b_t + \Delta b_t^{(\ell)}\Big)\,. \tag{10}$$

In this way, terrain encoding is conditioned on the robot motion state, while the body representation becomes progressively aware of terrain structure. Compared with one-shot late fusion, this design allows the policy to learn body–terrain coupling across multiple representation levels.

4) Global cross-attention fusion: After hierarchical interaction, a global body–terrain fusion is performed at the bottleneck stage. The final body feature forms the query, while the terrain bottleneck provides the key and value,

$$c_t = \mathrm{MHA}(W_q b_t,\; W_k z_t,\; W_v z_t). \tag{11}$$

The final control feature is then obtained by

$$f_t = f_{\mathrm{fuse}}([b_t,\, c_t]), \tag{12}$$

where $[\cdot, \cdot]$ denotes feature concatenation. The student action is predicted as

$$a_t = f_{\mathrm{act}}(f_t). \tag{13}$$

This stage summarizes terrain information after multi-level body–terrain interaction and produces the final feature for locomotion control.

C. Teacher-Guided Terrain Reconstruction

To improve robustness under corrupted terrain observations, we introduce a teacher-guided terrain reconstruction objective. In addition to predicting locomotion actions, the terrain branch is decoded to reconstruct a terrain target,

$$\tilde{e}_t = f_{\mathrm{dec}}(z_t, \{h_t^{(\ell)}\}_{\ell=0}^{L}), \tag{14}$$

where $f_{\mathrm{dec}}(\cdot)$ denotes a decoder with skip connections from the terrain encoder hierarchy.

The reconstruction target is obtained from the teacher-side observation stream. The reconstruction loss is defined as

$$\mathcal{L}_{\mathrm{rec}} = \big\|\tilde{e}_t - e_t^{\mathrm{T}}\big\|_2^2\,, \tag{15}$$

where e_t^{T} denotes the teacher-side terrain target used during training. This objective encourages the student to preserve terrain geometry that is useful for locomotion control, even when the input terrain observation is noisy or incomplete.

D. Stable Distillation Pipeline

The student policy is trained with teacher action supervision through the behavior distillation loss,

$$\mathcal{L}_{\mathrm{bc}} = \left\| a_t - a_t^{\mathrm{T}} \right\|_2^2 . \tag{16}$$

The overall training objective is

$$\mathcal{L} = \mathcal{L}_{\mathrm{bc}} + \lambda_{\mathrm{rec}} \mathcal{L}_{\mathrm{rec}}, \tag{17}$$

where λ_{rec} balances action distillation and terrain reconstruction.

To reduce the distribution mismatch between teacher-guided rollouts and student-driven deployment, we use Dataset Aggregation (DAgger)-style action mixing during rollout,

$$\bar{a}_t = \beta_k a_t^{\mathrm{T}} + (1 - \beta_k) a_t, \tag{18}$$

where β_k is annealed over training iterations. We further adopt replay-based optimization and staged training. Transitions collected during rollout are stored in a replay buffer and repeatedly sampled for mini-batch updates, which improves sample efficiency and reduces optimization variance. Before joint optimization, we first warm up the terrain encoder–decoder using the reconstruction objective, and then train the full student network with (17).

Finally, the student is trained under a structured corruption curriculum,

$$\hat{e}_t = \mathcal{C}_{\xi_t}(e_t), \tag{19}$$

where $\mathcal{C}_{\xi_t}$ denotes a corruption operator with progressively increasing difficulty. In our setting, the corruption includes noise, bias, drift, and spatially correlated perturbations. This curriculum improves robustness under imperfect terrain observations while maintaining stable optimization.

The overall training procedure of the proposed method is summarized in Algorithm 1.

As summarized in Algorithm 1, the proposed training procedure combines teacher action supervision, terrain reconstruction, and Dataset Aggregation (DAgger)-style rollout mixing to optimize the student policy under imperfect terrain perception. During rollout, mixed teacher–student actions are used for data collection, and the resulting transitions are stored in a replay buffer for mini-batch updates. In the student forward pass, proprioceptive history is encoded by the body stream, terrain observations are processed by the terrain stream, and hierarchical bidirectional cross-conditioning is applied to couple body dynamics with terrain structure across multiple representation levels. A global body–terrain fusion step is then used for action prediction, while the terrain branch is decoded to reconstruct the teacher-side terrain target. This training procedure improves robustness under corrupted terrain observations and supports reliable perceptive locomotion on challenging terrain.

Algorithm 1 BiCross-UNet Training Procedure

Require: Frozen teacher policy π_ϕ, student policy π_θ, replay buffer $\mathcal{B}$, mixing schedule $\{\beta_k\}_{k=1}^{K}$

Warm up the terrain encoder–decoder using $\mathcal{L}_{\mathrm{rec}}$
for $k = 1$ **to** K **do**
  for $t = 1$ **to** T **do**
    Obtain student observation $o_t = (x_{t-H+1:t}, \hat{e}_t)$ and teacher observation o_t^{T}
    Compute teacher action $a_t^{\mathrm{T}} = \pi_\phi(o_t^{\mathrm{T}})$ and student action $a_t = \pi_\theta(o_t)$
    Execute mixed action $\bar{a}_t = \beta_k a_t^{\mathrm{T}} + (1 - \beta_k) a_t$
    Step the environment and store $(o_t, o_t^{\mathrm{T}}, a_t^{\mathrm{T}}, e_t^{\mathrm{T}})$ in $\mathcal{B}$
  end for
  Sample a mini-batch from $\mathcal{B}$
  Encode proprioceptive history with the body stream and terrain observation with the terrain stream
  Apply hierarchical bidirectional cross-conditioning across all interaction levels
  Perform global body–terrain fusion to predict action $\hat{a}$
  Decode the terrain branch to reconstruct $\tilde{e}$
  Compute $\mathcal{L} = \mathcal{L}_{\mathrm{bc}} + \lambda_{\mathrm{rec}} \mathcal{L}_{\mathrm{rec}}$
  Update student parameters θ using $\nabla_\theta \mathcal{L}$
end for

IV. EXPERIMENTS

We evaluate the proposed *BiCross-UNet* from three aspects: 1) overall locomotion performance on challenging non-flat terrain, 2) robustness under noisy and imperfect terrain perception, and 3) the contribution of each component in the proposed architecture and training pipeline. We compare our method with two representative perceptive locomotion baselines, namely the attention-based map encoder in [1] and the teacher–student perceptive locomotion framework in [2]. We further provide ablation studies and qualitative results to analyze the proposed design.

A. Experimental Setup

a) Simulation platform and robot.: All experiments are conducted in Isaac Lab using a humanoid locomotion setup built on the G1 robot model. The policy receives proprioceptive observations together with a robot-centric terrain scan generated by a height scanner attached to the torso frame. The local terrain window covers a rectangular area in front of the robot and is discretized into a compact terrain feature vector as the exteroceptive input. The training environment is massively parallelized for large-scale data collection.

b) Terrain suite.: We evaluate the policy on a diverse set of non-flat terrains, including multi-step stairs, sparse discrete blocks, narrow gap-like structures, pit/high-platform terrains, wave-like uneven surfaces, and irregular rough terrains with local height discontinuities, as shown in Fig. 2. These terrains are selected to cover several representative failure modes in perceptive locomotion, including inaccurate foot placement on structured terrain, loss of stability under

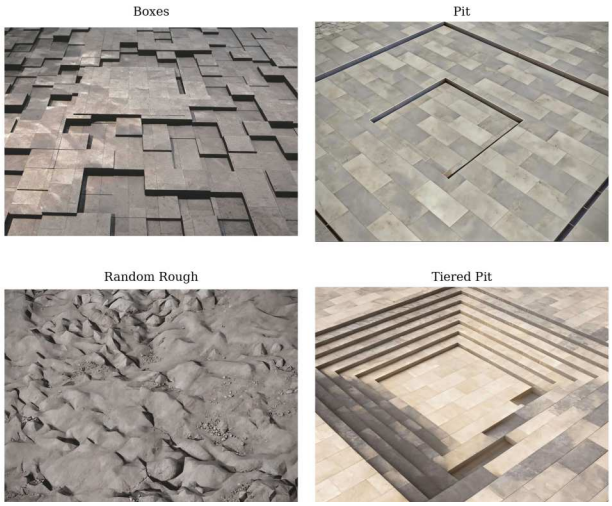


Fig. 2. Representative terrains in the evaluation benchmark. The terrain suite includes structured and irregular non-flat environments, such as stair-like layouts, sparse discrete footholds, gap/high-platform configurations, and rough uneven surfaces.

sustained elevation changes, and brittle behavior under sparse or corrupted terrain cues.

c) Training protocol and baselines.: For fair comparison, all methods are trained and evaluated under the same simulator, robot model, reward setting, command distribution, and terrain suite. Unless otherwise stated, only the policy architecture and training strategy are changed across methods. For the proposed method, the student is trained with corrupted terrain observations, while the teacher uses privileged clean terrain information. The full training pipeline includes Dataset Aggregation (DAgger)-style action mixing, replay-based optimization, staged training, and a structured terrain-corruption curriculum.

We compare against the following baselines:

- **AttentionMap** [1]: an attention-based perceptive locomotion policy that uses proprioception-conditioned terrain encoding for direct action generation.
- **HPC** [2]: a teacher–student perceptive locomotion framework trained with privileged supervision and student distillation.
- **BiCross-UNet (ours)**: the proposed method with hierarchical bidirectional cross-conditioning, global cross-attention fusion, teacher-guided terrain reconstruction, and the full distillation pipeline.

d) Evaluation metrics.: We evaluate each method using both task-level and control-level metrics. At the task level, we report the success rate, mean episode length, and fall rate. At the control level, we measure linear-velocity tracking error and yaw-rate tracking error:

$$e_v = \frac{1}{T}\sum_{t=1}^{T}\left\|\mathbf{v}_{xy,t}^{\mathrm{cmd}} - \mathbf{v}_{xy,t}\right\|_2, \tag{20}$$

$$e_\omega = \frac{1}{T}\sum_{t=1}^{T}\left|\omega_{z,t}^{\mathrm{cmd}} - \omega_{z,t}\right|. \tag{21}$$

TABLE I

QUANTITATIVE COMPARISON ON THE CHALLENGING-TERRAIN BENCHMARK. HIGHER SUCCESS RATE AND LONGER EPISODE LENGTH ARE BETTER, WHILE LOWER FALL RATE AND TRACKING ERRORS ARE BETTER.

Method	**Succ.** ↑	**Ep. Len.** ↑	**Fall** ↓	e_v ↓	e_ω ↓
AttentionMap [1]	68.4	12.7	24.6	0.214	0.318
HPC [2]	76.9	15.9	16.8	0.182	0.271
BiCross-UNet (ours)	**84.7**	**18.6**	**10.5**	**0.149**	**0.224**

Here, $\mathbf{v}_{xy,t}^{\mathrm{cmd}}$ and $\omega_{z,t}^{\mathrm{cmd}}$ denote the commanded planar velocity and yaw rate, and $\mathbf{v}_{xy,t}$ and $\omega_{z,t}$ denote the measured robot velocities. A trial is counted as successful if the robot traverses the assigned terrain segment without falling before the episode terminates. Unless otherwise stated, all reported results are averaged over 3 random seeds and 100 evaluation episodes per method. For robustness evaluation, we additionally report performance under structured terrain corruption, including additive noise, drift, bias distortion, and spatially correlated perturbations.

B. Main Quantitative Results

Table I reports the quantitative comparison on the challenging-terrain benchmark. Overall, *BiCross-UNet* achieves the best performance among the compared methods. In particular, the proposed method obtains the highest success rate and the longest mean episode length, while also reducing the fall rate and command-tracking errors. Compared with **AttentionMap**, the proposed method shows clearer advantages on sparse and highly structured terrains, indicating that repeated body–terrain interaction during encoding is more effective than one-shot terrain fusion. Compared with **HPC**, *BiCross-UNet* yields stronger overall locomotion performance under the same evaluation protocol, which suggests that the proposed cross-conditioned representation and teacher-guided terrain reconstruction improve both terrain-aware control and robustness under imperfect terrain perception.

Across terrain categories, the proposed method maintains better stability on stair and high-platform terrains, produces more reliable traversal on sparse discrete and gap-like terrains, and achieves lower tracking error on wave-like and irregular rough terrains. These results indicate that the proposed method improves both terrain-aware control and robustness under degraded terrain perception.

C. Robustness to Imperfect Terrain Perception

To explicitly evaluate robustness, we test all methods under increasing terrain-perception degradation. During evaluation, the terrain input is corrupted using perturbations consistent with the training setting, including random noise, drift, bias distortion, and spatially correlated perturbations.

Table II shows that *BiCross-UNet* degrades more gracefully than both baselines as terrain observations become less reliable. The performance gap becomes larger under strong structured corruption, where local terrain errors accumulate spatially rather than appearing as independent noise. This

TABLE II

ROBUSTNESS UNDER STRUCTURED TERRAIN CORRUPTION. WE REPORT SUCCESS RATE AND COMMAND-TRACKING ERRORS UNDER INCREASING CORRUPTION INTENSITY. HIGHER SUCCESS RATE IS BETTER, WHILE LOWER TRACKING ERRORS ARE BETTER.

Method	**Success (%) ↑**				e_v ↓				e_ω ↓			
	0%	**50%**	**100%**	**200%**	**0%**	**50%**	**100%**	**200%**	**0%**	**50%**	**100%**	**200%**
AttentionMap [1]	68.4	59.1	44.7	28.9	0.214	0.249	0.312	0.397	0.318	0.367	0.451	0.576
HPC [2]	76.9	71.8	63.2	48.6	0.182	0.206	0.247	0.318	0.271	0.301	0.354	0.438
BiCross-UNet (ours)	**84.7**	**81.6**	**75.4**	**66.9**	**0.149**	**0.163**	**0.184**	**0.221**	**0.224**	**0.243**	**0.271**	**0.329**

result is consistent with the proposed design. Hierarchical bidirectional cross-conditioning enables repeated interaction between body state and terrain features, while teacher-guided terrain reconstruction encourages the latent terrain representation to preserve useful geometric structure under corrupted observations.

D. Ablation Studies

To evaluate the contribution of each component, we conduct the following ablations:

- **w/o BiCross**: remove hierarchical bidirectional cross-conditioning and retain only late-stage fusion.
- **w/o GlobalAttn**: remove the global cross-attention fusion module.
- **w/o Recon**: remove the teacher-guided terrain reconstruction objective.
- **w/o Curriculum**: remove the structured corruption curriculum during student training.
- **w/o Replay+DAgger**: remove replay-based updates and DAgger-style action mixing.

Table III reports the contribution of each component in the proposed method. Removing *BiCross* causes the largest performance drop, which confirms the importance of repeated body–terrain interaction during encoding. Removing *GlobalAttn* further degrades tracking quality, indicating that high-level terrain summarization remains useful after hierarchical interaction. Removing *Recon* reduces robustness and overall locomotion performance, showing that action imitation alone is insufficient to learn a stable terrain representation under corrupted observations. Finally, removing the curriculum or the replay-based distillation strategy leads to weaker robustness and less stable control, which validates the necessity of the full training pipeline.

TABLE III

ABLATION STUDY OF THE PROPOSED METHOD ON THE CHALLENGING-TERRAIN BENCHMARK. HIGHER SUCCESS RATE AND LONGER EPISODE LENGTH ARE BETTER, WHILE LOWER FALL RATE AND TRACKING ERRORS ARE BETTER.

Method	**Succ. ↑**	**Ep. Len. ↑**	**Fall ↓**	e_v ↓	e_ω ↓
w/o BiCross	75.9	15.1	18.2	0.186	0.281
w/o GlobalAttn	79.8	16.8	14.4	0.169	0.249
w/o Recon	77.1	15.9	16.0	0.178	0.268
w/o Curriculum	78.4	16.1	15.7	0.173	0.258
w/o Replay+DAgger	77.6	15.8	16.6	0.180	0.271
BiCross-UNet (full)	**84.7**	**18.6**	**10.5**	**0.149**	**0.224**

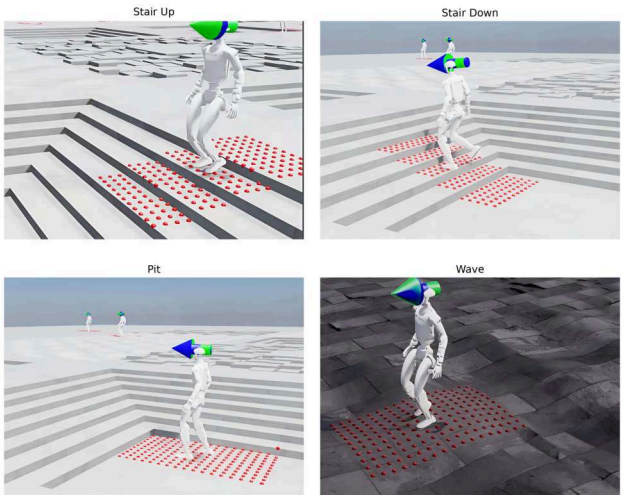


Fig. 3. Representative qualitative rollouts in Isaac simulation. The humanoid robot is evaluated on stair-like, pit/high-platform, and wave-like terrains. The red points indicate the perceived terrain region used by the policy.

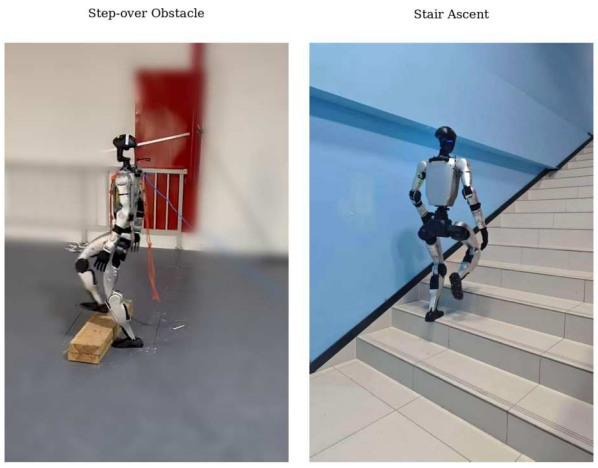


Fig. 4. Representative real-world experiments. The humanoid robot performs stair ascent and obstacle crossing in the real environment, demonstrating successful sim-to-real transfer of the learned policy.

E. Qualitative Analysis

Fig. 3 shows representative rollouts in Isaac simulation. The proposed method maintains stable body posture and consistent stepping behavior on stair-like, pit/high-platform,

and wave-like terrains. Fig. 4 further presents real-world experiments including stair ascent and obstacle crossing, suggesting that the learned policy transfers to hardware and remains effective under deployment conditions.

To better understand the learned representation, we also inspect the terrain-attention behavior in challenging stair scenes. Qualitative inspection suggests that the policy focuses on geometrically informative regions near step transitions and feasible landing areas, rather than distributing attention uniformly over the entire local map. This observation is consistent with the motivation of the proposed design, namely that terrain perception should be conditioned on body state, and body control should in turn be refined by terrain context.

V. CONCLUSION

In this paper, we proposed *BiCross-UNet*, a robust perceptive locomotion framework for humanoid robots on challenging terrain under imperfect terrain perception. By combining hierarchical bidirectional cross-conditioning, teacher-guided terrain reconstruction, and a stable distillation pipeline, the proposed method improves both terrain-aware control and robustness under corrupted terrain observations. Experimental results showed that *BiCross-UNet* outperforms representative perceptive locomotion baselines on challenging non-flat terrains and degrades more gracefully under structured terrain corruption. Future work will focus on more challenging real-world perception conditions and broader humanoid locomotion scenarios.

REFERENCES

[1] J. He, C. Zhang, F. Jenelten, R. Grandia, M. Bächer, and M. Hutter, "Attention-based map encoding for learning generalized legged locomotion," *Science Robotics*, vol. 10, no. 105, p. eadv3604, 2025.

[2] W. Sun, B. Cao, L. Chen, Y. Su, Y. Liu, Z. Xie, and H. Liu, "Learning perceptive humanoid locomotion over challenging terrain," *arXiv preprint arXiv:2503.00692*, 2025.

[3] J. Hwangbo, J. Lee, A. Dosovitskiy, D. Bellicoso, V. Tsounis, V. Koltun, and M. Hutter, "Learning agile and dynamic motor skills for legged robots," *Science Robotics*, vol. 4, no. 26, p. eaau5872, 2019.

[4] J. Lee, J. Hwangbo, L. Wellhausen, V. Koltun, and M. Hutter, "Learning quadrupedal locomotion over challenging terrain," *Science Robotics*, vol. 5, no. 47, p. eabc5986, 2020.

[5] N. Rudin, D. Hoeller, P. Reist, and M. Hutter, "Learning to walk in minutes using massively parallel deep reinforcement learning," in *Proceedings of the 5th Conference on Robot Learning*, 2022, pp. 91–100.

[6] T. Miki, J. Lee, J. Hwangbo, L. Wellhausen, V. Koltun, and M. Hutter, "Learning robust perceptive locomotion for quadrupedal robots in the wild," *Science Robotics*, vol. 7, no. 62, p. eabk2822, 2022.

[7] F. Jenelten, J. He, F. Farshidian, and M. Hutter, "Dtc: Deep tracking control," *Science Robotics*, vol. 9, no. 86, p. eadh5401, 2024.

[8] J. Long, J. Ren, M. Shi, Z. Wang, T. Huang, P. Luo, and J. Pang, "Learning humanoid locomotion with perceptive internal model," in *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, 2025.

[9] H. Wang, Z. Wang, J. Ren, Q. Ben, T. Huang, W. Zhang, and J. Pang, "Beamdojo: Learning agile humanoid locomotion on sparse footholds," in *Proceedings of Robotics: Science and Systems*, 2025.

[10] I. M. A. Nahrendra, B. Yu, and H. Myung, "Dreamwaq: Learning robust quadrupedal locomotion with implicit terrain imagination via deep reinforcement learning," in *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, 2023, pp. 5078–5084.

[11] X. Gu, Y.-J. Wang, X. Zhu, C. Shi, Y. Guo, Y. Liu, and J. Chen, "Advancing humanoid locomotion: Mastering challenging terrains with denoising world model learning," in *Proceedings of Robotics: Science and Systems*, 2024.

[12] S. Tan, Z. Hou, Z. Zhang, L. Xu, M. Zhang, Z. He, C. Xu, F. Gao, and Y. Cao, "Real-time spatial-temporal traversability assessment via feature-based sparse gaussian process," in *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025, pp. 17 533–17 540.

[13] L. Xu, K. Chai, Z. Han, H. Liu, C. Xu, Y. Cao, and F. Gao, "An efficient trajectory planner for car-like robots on uneven terrain," in *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2023, pp. 2853–2860.

[14] X. Li, L. Xu, X. Huang, D. Xue, Z. Zhang, Z. Han, C. Xu, Y. Cao, and F. Gao, "SEB-Naver: A SE(2)-based local navigation framework for car-like robots on uneven terrain," *arXiv preprint arXiv:2503.02412*, 2025.

[15] L. Xu, K. Chai, B. An, S. Ji, Z. Hou, J. Gan, Q. Wang, Y. Zhou, X. Li, J. Lin, Z. Han, C. Xu, Y. Cao, and F. Gao, "Tracailer: An efficient trajectory planner for tractor-trailer robots in unstructured environments," *IEEE Transactions on Automation Science and Engineering*, vol. 22, pp. 21 118–21 138, 2025.

[16] S. Tan, Z. He, K. Zhang, Z. Hou, Z. Wang, and M. Wang, "From gaussian processes to ellipsoidal obstacles: Safety control barrier functions and model predictive control," in *Proceedings of the 30th International Conference on Automation and Computing (ICAC)*, 2025, pp. 1–7.

[17] Q. He, Z. Hou, and X. Zhu, "A novel algorithm for ship route planning considering motion characteristics and ENC vector maps," *Journal of Marine Science and Engineering*, vol. 11, no. 6, p. 1102, 2023.

[18] L. Hu, Z. Li, X. Zhu, D. Li, and R. Song, "DPR-Splat: Depth and pose refinement with sparse-view 3d gaussian splatting for novel view synthesis," in *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025, pp. 12 900–12 907.

[19] L. Hu, N. Ait Oufroukh, F. Bonardi, and R. Ghandour, "EC3R-SLAM: Efficient and consistent monocular dense SLAM with feed-forward 3d reconstruction," *arXiv preprint arXiv:2510.02080*, 2025.